Body fat and animal protein intakes are associated with adrenal androgen secretion in children.

BACKGROUND: Adrenarche is the increase in adrenal androgen (AA) production starting in childhood. Until now, it has been unknown whether or not nutritional factors modulate adrenarche. OBJECTIVE: The objective was to examine whether body composition and certain dietary intakes are associated with AA production in children after accounting for urinary indicators of major adrenarche-related steroidogenic enzymes. DESIGN: Androgen and glucocorticoid metabolites were profiled by gas chromatography-mass spectrometry in 24-h urine samples of 137 healthy prepubertal children aged 3-12 y, for whom birth characteristics, growth velocity data, and 3-d weighed-diet record information were available. Associations of the sum of C19 metabolites (reflecting daily AA secretion) with nutritional factors [fat mass (FM), fat-free mass (FFM), nutrient intakes, glycemic index, and glycemic load] and AA-relevant estimates of steroidogenic enzyme were examined in stepwise multiple regression models adjusted for age, sex, urine volume, and total energy intake. Enzyme activity estimates were calculated by using specific urinary steroid metabolite ratios. RESULTS: Of the nutrition-relevant predictors, FM ($P < 0.0001$) explained most of the variation of AA secretion ($R(2) = 5\%$). Animal protein intake was also positively associated with AA secretion ($P < 0.05$), which explained 1% of its variation. FFM ($P = 0.1$) and total protein intake ($P = 0.05$) showed positive trends. The difference in daily AA secretion between the lowest and highest quartile of FM was comparable to that between the lowest and highest estimated activity of one of the major steroidogenic enzymes. CONCLUSIONS: Body fat mass may relevantly influence prepubertal adrenarchal androgen status. In addition, animal protein intake may also make a small contribution to AA secretion in children.